

RG05

Soils of Rajasthan

One to two questions a paper, and the agro-climatic zones are the more reliable of the two halves - RPSC has asked how many zones there are, and which zone a named group of districts belongs to. Soil questions come as soil-to-district or soil-to-order matching.

Soil in Rajasthan is really a map question. Each soil type belongs to a block of districts, and that block is decided by rock, rainfall and river. Attached to it is the other half of the chapter - the ten agro-climatic zones, which RPSC has asked by number, by name and by district. Learn the two tables in this chapter and you have the whole thing.

Soils and Agro-climatic Zones of Rajasthan

Soil types

- Desert / arid sandy - largest area
- Brown (non-calcic) - Luni basin
- Red-yellow - Aravalli granite belt
- Red loamy - humid south
- Mixed red-black
- Black / medium black - Hadoti
- Alluvial - eastern plain
- Saline-alkaline (Reh, Kallar, Usar)
- Lithosols and regosols - Aravalli slopes

USDA / ICAR orders

- Entisol - largest, western Rajasthan
- Aridisol - true desert core
- Inceptisol - southern Aravalli, Mewar, Vagad
- Vertisol - Hadoti black cotton soil
- Alfisol - eastern plain

Soil problems

- Wind erosion and dune movement - west
- Ravine and gully erosion - Chambal tract
- Waterlogging - IGNP command
- Secondary salinity and alkalinity
- Largest degraded area of any Indian state

Reclamation

- Gypsum for alkaline soil
- Sub-surface drainage for waterlogging
- Shelterbelts and dune stabilisation
- Contour bunding and check dams

Agro-climatic zones

- Ten zones in five regions I to V
- I-A, I-B, I-C - arid west
- II-A, II-B - transitional plains
- III-A, III-B - eastern plains
- IV-A, IV-B - southern plains
- V - Hadoti, most productive

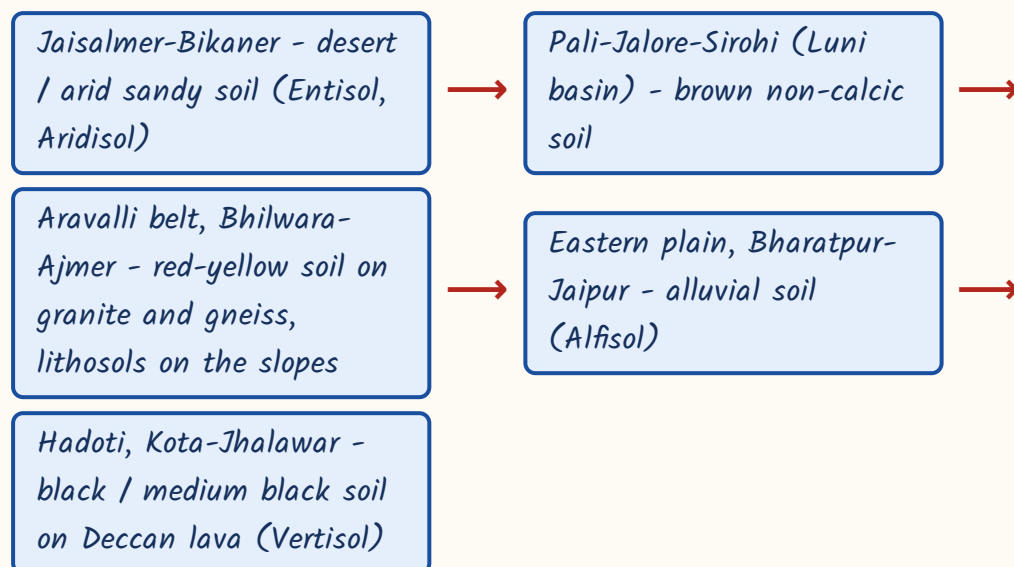
Why Rajasthan's soils are what they are

Soil is weathered rock plus organic matter plus water. In Rajasthan the rock is granite and gneiss in the Aravallis, Deccan basalt in Hadoti, and Vindhyan sandstone in the south-east; the water is scarce; and the organic matter is scarcer still because there is not enough vegetation to make it. That is why the state's soils are, as a rule, poor in humus and nitrogen. Move west and the wind

takes over as the depositing agent, so you get sand. Move east and rivers take over, so you get alluvium.

- Parent material decides colour and texture - basalt gives black, granite and gneiss give red-yellow, wind-blown sand gives desert soil, river deposition gives alluvium.
- Rainfall decides depth and leaching - shallow, calcareous and unleached in the west; deeper and more developed in the east and south.
- Almost every soil group in Rajasthan is deficient in nitrogen and organic matter; phosphorus is often deficient too.
- Only alluvial and black soils are naturally productive without heavy input.

Soil transect - walk from Jaisalmer to Jhalawar



The soil types with their districts - the master table

This is the table that answers the matching question. Read it as blocks of the map, not as isolated names: sand in the west, brown along the Luni, red in the Aravallis and the south, alluvium in the east, black in Hadoti, salt around the canal command and the salt lakes.

Soil types of Rajasthan and their districts

Soil type	Districts	Character
Desert / arid sandy	Jaisalmer, Barmer, Bikaner, Jodhpur, Churu, Nagaur	Largest area in the state; low humus and nitrogen; calcareous; highly porous
Brown (non-calcic brown)	Luni basin - Pali, Jalore, Sirohi, Ajmer, parts of Sikar and Jhunjhunu	Comparatively rich in phosphate
Red-yellow	Sawai Madhopur, Bhilwara, Udaipur, Rajsamand, Ajmer, Tonk	From granite and gneiss; deficient in nitrogen and calcium
Red loamy	Banswara, Dungarpur, Pratapgarh, Udaipur, Chittorgarh	Humid south; iron oxides give the colour
Mixed red and black	Chittorgarh, Pratapgarh, Bhilwara, Banswara, Jhalawar	Where Aravalli metamorphics meet Deccan lava; potash sufficient, N and P deficient
Black / medium black	Hadoti - Kota, Bundi, Baran, Jhalawar	From Deccan lava; high clay, strong moisture retention
Alluvial	Bharatpur, Dholpur, Alwar, Jaipur, Dausa, Tonk, Karauli, Sawai Madhopur, Ganganagar canal tract	The most fertile group in the state
Saline-alkaline (Reh, Kallar, Usar)	Sri Ganganagar, Hanumangarh, Barmer, Jalore, Pali, Sambhar-Pachpadra belt	From capillary rise of salts and over-irrigation
Lithosols and regosols	Aravalli slopes	Thin skeletal soils, almost no profile development, very low agricultural value

Points that turn up as statement questions

- Desert soils are highly porous with low water-holding capacity, poor in humus and nitrogen, but calcareous and often carrying a kankar or gypsum layer.
- Black soils crack deeply in summer and are self-mulching - the crack lets air in and closes when wet.

- Alluvial soils of the eastern plain are laid down by the Banganga, Gambhir, Chambal and Banas systems.
- Red-yellow soil comes from granite and gneiss weathering, black from Deccan basalt, desert soil from wind-blown sand, alluvium from river deposition.

TRAP

Two set traps. Churu is arid Thar - it carries DESERT SANDY soil, never red loam. And black soil belongs to HADOTI, not to Shekhawati; Shekhawati carries sandy and brown soils.

The soil orders - USDA and ICAR taxonomy

The modern classification names soils by order rather than by colour. Rajasthan has five orders. Map each order onto the colour-name soil you already know and you never have to learn the districts twice. Entisol is the youngest soil with no horizon development; Aridisol is the true desert order; Vertisol is the swelling black clay; Alfisol is the moderately leached soil with a clay-enriched subsoil; Inceptisol is the young but slightly developed soil in between.

Soil orders of Rajasthan mapped to regions

Order	What it is	Districts / region
Entisol	Youngest order, almost no horizon development - LARGEST extent in the state	Western Rajasthan generally
Aridisol	The true desert order	Jaisalmer, Bikaner, Churu, Jhunjhunu
Inceptisol	Young soils with the beginnings of horizon development	Udaipur, Rajsamand, Chittorgarh, Banswara, Dungarpur

Order	What it is	Districts / region
Vertisol	Deep swelling clays, self-mulching, crack widely when dry - the black cotton soil	Kota, Bundi, Baran, Jhalawar (Hadoti)
Alfisol	Moderately leached, clay-enriched subsoil	Alwar, Bharatpur, Dholpur, Karauli

The two orders RPSC contrasts most - Entisol versus Vertisol

Entisol	Vertisol
Largest extent in Rajasthan	Confined to the Hadoti plateau
Western arid plain	Kota, Bundi, Baran, Jhalawar
Youngest order - no horizon development	Deep, mature swelling-clay profile
Sandy, porous, very low moisture retention	Heavy clay, excellent moisture retention
Corresponds to the desert sandy soil	Corresponds to the black cotton soil

YAAD RAKHO

Yaad rakho by the name's meaning - ENTisol from 'recent', so youngest and biggest here; ARIDisol for the arid core; VERTisol from 'vertere', to turn, because the cracking clay churns itself; ALFisol has Aluminium and iron (Al-Fe) in a clay-rich B horizon; INCEPTisol is at the inception of development.

TRAP

Oxisols do NOT occur in Rajasthan. They are the deeply weathered soils of the humid equatorial tropics. Rajasthan's five are Entisol, Aridisol, Inceptisol, Vertisol and Alfisol.

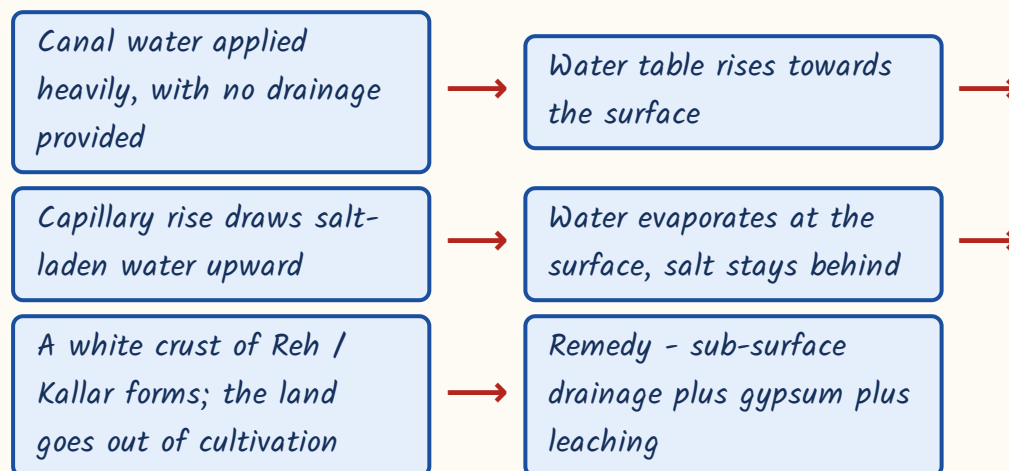
Soil problems - each one belongs to a region

Every soil problem in Rajasthan has an address. Wind erosion belongs to the Thar, ravines to the Chambal tract, waterlogging to the canal command, salinity to the canal command and the salt-lake belt. If you can name the region, you can answer the question - and this is exactly how the matching question is set.

Soil problem matched to the area affected

Problem	Area chiefly affected	Cause
Wind erosion and dune movement	Western Rajasthan - Jaisalmer, Barmer, Bikaner, Jodhpur	High wind velocity, loose sand, sparse vegetation
Sheet, rill and gully erosion (ravines)	Chambal tract - Dholpur, Karauli, Sawai Madhopur	Steep gradients and torrential runoff on soft alluvium
Waterlogging	Indira Gandhi Nahar command - Sri Ganganagar, Hanumangarh	Over-irrigation with inadequate drainage
Secondary salinity and alkalinity	IGNP command; also Sambhar and Pachpadra surroundings	Rising water table bringing salts to the surface

How a canal command turns saline - the sequence RPSC likes



- The Chambal ravines of Dholpur, Karauli and Sawai Madhopur are among India's most severely gullied badlands.
- Rajasthan has the largest area under land degradation and desertification of any Indian state, as mapped in ISRO's Desertification and Land Degradation Atlas.

- Wind erosion is the principal process of land degradation in the state.
- Reh, Kallar and Usar are the local names for salt-affected land.

Reclamation – the fix for each problem

Match the cure to the disease. Gypsum for alkalinity, drainage for waterlogging, plantation for wind, engineering for water erosion. RPSC has asked which amendment reclaims an alkaline soil – the answer is gypsum, and lime is the distractor.

- Gypsum for alkaline soils – the calcium in gypsum displaces sodium from the soil complex, and the sodium salts are then leached out.
- Sub-surface drainage for waterlogging in the IGNP command.
- Shelterbelt plantation and sand-dune stabilisation against wind erosion.
- Contour bunding, check dams and gully plugging against water erosion in the ravine tract.

Do not confuse the two amendments. Gypsum is for alkaline soil, which is Rajasthan's problem. Lime is for acidic soil, which is not a Rajasthan problem at all. If an option offers lime for reclaiming a Rajasthan soil, it is wrong – that single swap is the whole question.

The ten agro-climatic zones – a classic RPSC target

Rajasthan is divided into ten agro-climatic zones. They were delineated under the National Agricultural Research Project on the basis of rainfall, soil, temperature and topography, and they are grouped into five broad regions numbered I to V. The numbering runs from the arid north-west to the humid south-east, so the zone number and the rainfall rise together. Learn the table;

then learn the shape of the numbering, which is what lets you reconstruct it under pressure.

The ten agro-climatic zones of Rajasthan

Zone	Name	Core districts
I-A	Arid Western Plain	Barmer and part of Jodhpur; rainfall about 200-370 mm; sandy soil
I-B	Irrigated North-Western Plain	Sri Ganganagar and Hanumangarh - the only fully canal-irrigated zone; wheat, cotton, mustard
I-C	Hyper Arid Partial Irrigated Zone	Bikaner, Churu and Jaisalmer - the driest zone of the state
II-A	Transitional Plain of Inland Drainage	Sikar, Jhunjhunu, Nagaur and part of Churu - the Shekhawati tract
II-B	Transitional Plain of Luni Basin	Jalore, Pali and parts of Sirohi and Jodhpur
III-A	Semi-Arid Eastern Plain	Jaipur, Ajmer, Dausa and Tonk; rainfall roughly 500-600 mm
III-B	Flood Prone Eastern Plain	Alwar, Bharatpur, Dholpur, Karauli and Sawai Madhopur
IV-A	Sub-Humid Southern Plain	Udaipur, Chittorgarh, Rajsamand, Bhilwara and part of Pratapgarh
IV-B	Humid Southern Plain	Dungarpur and Banswara - among the highest-rainfall zones
V	Humid South-Eastern Plain (Hadoti)	Kota, Bundi, Baran and Jhalawar - deep black soils, the most productive farmland of the state

YAAD RAKHO

Yaad rakho the shape 3-2-2-2-1. Region I has THREE zones (the arid west: Arid, Irrigated, Hyper-arid), II has TWO (both Transitional - Inland Drainage and Luni Basin), III has TWO (both Eastern - Semi-Arid and Flood Prone), IV has TWO (both Southern - Sub-Humid and Humid), and V stands ALONE (Hadoti). $3+2+2+2+1 = 10$. Recite the shape first, then fill in the districts.

- A second aid - the numbers climb with the rain: Region I is driest, Region V is wettest.
- The sequence runs north-west to south-east, the same diagonal as the rainfall chapter.
- So if you know the rainfall map, you already know the order of the zones.

ASKED IN RAS

Asked directly by RPSC - into how many agro-climatic zones has Rajasthan been divided? Answer: ten, grouped into five broad regions numbered I to V.

ASKED IN RAS

Asked - Bikaner, Churu and Jaisalmer fall in which agro-climatic zone?

Answer: Zone I-C, the Hyper Arid Partial Irrigated Zone, the driest zone of the state.

ASKED IN RAS

Asked - Jaipur, Ajmer, Dausa and Tonk fall in which zone? Answer: Zone III-A, the Semi-Arid Eastern Plain. Its companion, Zone III-B Flood Prone Eastern Plain, covers Alwar, Bharatpur, Dholpur, Karauli and Sawai Madhopur.

TRAP

The most common wrong pairing is putting Bikaner and Churu into Zone IV-A. Zone IV-A is the Sub-Humid SOUTHERN Plain - Udaipur, Chittorgarh, Rajsamand, Bhilwara. Bikaner and Churu are hyper-arid Zone I-C. Also do not slide Alwar-Bharatpur-Dholpur into II-B; II-B is the Luni basin.

- Zone I-B has low natural rainfall - its prosperity rests entirely on canal irrigation, which is why the name says 'Irrigated'.
- Ascending rainfall order across zones: I-C, then II-A, then III-A, then IV-B.
- Zones I-A, I-B and I-C together make up the arid western plain region; IV-A, IV-B and V cover the humid south and south-east.
- Zone V is the state's most productive farmland - deep black Vertisols with strong moisture retention.

One caution about district names. The zones were framed on the older district map. Rajasthan reorganised its districts in 2023 and revised that again in December 2024, and the state now has 41 districts and 7 divisions. Newer districts sit inside their old parents - Balotra inside the Barmer tract of Zone I-A, Phalodi inside the Jodhpur tract, Beawar and Kekri's area inside the Ajmer tract of Zone III-A, Didwana-Kuchaman inside the Nagaur tract of Zone II-A, Deeg and Khairthal-Tijara inside the Bharatpur and Alwar tracts of Zone III-B, Salumbar inside the Udaipur tract of Zone IV-A. RPSC still sets these questions in the old district names - answer in the names the question uses.

Soil health infrastructure and the current file

The administrative half of this chapter is short and easy marks. The state Department of Agriculture runs soil testing laboratories, static and mobile, in

the districts, and issues Soil Health Cards to farmers with the nutrient status of their own field and a fertiliser recommendation.

- Soil Health Card scheme - the card carries the field's nutrient status and a crop-wise fertiliser recommendation.
- Soil testing laboratories, both static and mobile, operate at district level under the state Department of Agriculture.
- The purpose is to correct Rajasthan's chronic nitrogen and organic-matter deficiency without over-applying urea.

CURRENT LINK

Current link (as of September 2026): the Soil Health Card scheme completed a decade in 2025 and is now folded into the wider soil-health component of central agriculture schemes. On the degradation side, ISRO's Desertification and Land Degradation Atlas continues to place Rajasthan first among Indian states for degraded area - that ranking is the fact most likely to appear in a current-affairs or environment question.

Rapid linkages - be able to recite these both ways

Before the exam, test yourself in both directions. Given the district, name the soil, the order and the zone. Given the zone, name the districts. These three-way links are what the matching questions test.

District block - soil - order - zone, in one line each

District block	Soil and order	Agro-climatic zone
Jaisalmer, Bikaner, Churu	Desert sandy; Aridisol / Entisol	I-C Hyper Arid Partial Irrigated
Sri Ganganagar, Hanumangarh	Alluvial but salt-affected; Entisol	I-B Irrigated North-Western Plain

District block	Soil and order	Agro-climatic zone
Sikar, Jhunjhunu, Nagaur	Sandy and brown; Entisol / Aridisol	II-A Transitional Plain of Inland Drainage
Pali, Jalore, Sirohi	Brown non-calcic; Entisol	II-B Transitional Plain of Luni Basin
Jaipur, Ajmer, Dausa, Tonk	Red-yellow and alluvial; Alfisol	III-A Semi-Arid Eastern Plain
Alwar, Bharatpur, Dholpur, Karauli	Alluvial; Alfisol	III-B Flood Prone Eastern Plain
Udaipur, Chittorgarh, Rajsamand, Bhilwara	Red-yellow and mixed red-black; Inceptisol	IV-A Sub-Humid Southern Plain
Dungarpur, Banswara	Red loamy; Inceptisol	IV-B Humid Southern Plain
Kota, Bundi, Baran, Jhalawar	Black / medium black; Vertisol	V Humid South-Eastern Plain (Hadoti)

- The soil order and the zone move together across the map - that is the shortcut.
- Aridisol with Zone I; Entisol with Zones I and II; Alfisol with Zone III.
- Inceptisol with Zone IV; Vertisol with Zone V.
- One diagonal, three labels - district, soil order, agro-climatic zone.

REVISION RECAP

1. Desert / arid sandy soil covers the largest area - Jaisalmer, Barmer, Bikaner, Jodhpur, Churu, Nagaur; porous, calcareous, poor in humus and nitrogen.
2. Brown non-calcic soil = Luni basin (Pali, Jalore, Sirohi, Ajmer), comparatively rich in phosphate.
3. Red-yellow soil = Sawai Madhopur, Bhilwara, Udaipur, Rajsamand, Ajmer, Tonk, on granite and gneiss.
4. Red loamy = Banswara, Dungarpur, Pratapgarh, Udaipur, Chittorgarh; mixed red-black = Chittorgarh, Pratapgarh, Bhilwara, Banswara, Jhalawar.
5. Black / medium black = Hadoti (Kota, Bundi, Baran, Jhalawar), from Deccan lava, high clay, strong moisture retention.

6. Alluvial soil of the eastern plain is the most fertile group; saline-alkaline (Reh, Kallar, Usar) afflicts Sri Ganganagar-Hanumangarh and the Sambhar-Pachpadra belt.
7. Lithosols and regosols are the thin skeletal soils of the Aravalli slopes.
8. Orders: Entisol largest (west), Aridisol (Jaisalmer, Bikaner, Churu, Jhunjhunu), Inceptisol (Udaipur-Vagad), Vertisol (Hadoti), Alfisol (eastern plain). No Oxisols in Rajasthan.
9. Vertisol = black cotton soil, self-mulching, deep summer cracks; Entisol = youngest, no horizons.
10. Wind erosion and dunes = the west; ravines and gullies = the Chambal tract of Dholpur, Karauli, Sawai Madhopur.
11. Waterlogging and secondary salinity = the Indira Gandhi Nahar command in Sri Ganganagar and Hanumangarh.
12. Rajasthan has the largest degraded and desertified area of any Indian state (ISRO Desertification and Land Degradation Atlas).
13. Reclamation: gypsum for alkalinity, sub-surface drainage for waterlogging, shelterbelts and dune stabilisation for wind, contour bunding and check dams for water erosion. Lime is for acidic soil, not here.
14. Rajasthan has TEN agro-climatic zones grouped into five regions I to V, delineated under the National Agricultural Research Project.
15. Shape of the grouping: 3-2-2-2-1 across regions I to V; zone number and rainfall rise together from north-west to south-east.
16. I-A Arid Western Plain (Barmer, part Jodhpur); I-B Irrigated North-Western Plain (Sri Ganganagar, Hanumangarh); I-C Hyper Arid (Bikaner, Churu, Jaisalmer).
17. II-A Inland Drainage (Sikar, Jhunjhunu, Nagaur); II-B Luni Basin (Jalore, Pali, part Sirohi and Jodhpur).
18. III-A Semi-Arid Eastern Plain (Jaipur, Ajmer, Dausa, Tonk); III-B Flood Prone Eastern Plain (Alwar, Bharatpur, Dholpur, Karauli, Sawai Madhopur).
19. IV-A Sub-Humid Southern Plain (Udaipur, Chittorgarh, Rajsamand, Bhilwara); IV-B Humid Southern Plain (Dungarpur, Banswara); V Hadoti (Kota, Bundi, Baran, Jhalawar), the most productive.
20. The zones were framed on the pre-2023 district map; Rajasthan now has 41 districts and 7 divisions after the December 2024 revision.